

TASK 1: INTRODUCTION

Page 1.

SIEM - The Core of SOC Visibility & Detection

- In a **SOC** (Security Operations Center), analysts receive thousands of logs from many devices. These logs tell us what is happening across the network.
- A **SIEM** solution is used to collect, normalize, store, correlate and analyze those logs in **ONE** place.

Q. What does SIEM stand for?

SIEM stands for
Security Information and Event Management

LEARNING OBJECTIVES

- 1 Understand the different types of log sources
- 2 Identify the limitations of working with isolated logs
- 3 Recognize the importance of a SIEM solution
- 4 Explore the features of a SIEM solution
- 5 Learn various types of log sources and their ingestion in the SIEM
- 6 Understand the process behind alerting and alert analysis

WHAT IS SIEM?

A SIEM (Security Information and Event Management) is a centralized platform that collects logs from various sources, normalizes the data, correlates events, detects threats, and provides visibility across the entire IT environment.



Think of SIEM
as the "Brain"
of the SOC.

WHY IS IT NEEDED?

- Networks generate huge amounts of logs every second.
- Attackers try to hide by spreading activity across systems.
- Without a central solution, critical threats can be missed.
- SIEM helps in early detection, investigation & response.



Page 3.

(3) IMPORTANCE OF A SIEM SOLUTION

- ✓ Centralized visibility across the entire environment
- ✓ Correlates events to detect advanced threats
- ✓ Real-time alerting for quick response
- ✓ Historical data for investigations & forensics
- ✓ Compliance reporting (PCI-DSS, ISO, HIPAA, etc.)
- ✓ Helps in threat hunting & trend analysis

SIEM turns
raw logs
into
actionable
intelligence!

(4) FEATURES OF A SIEM SOLUTION

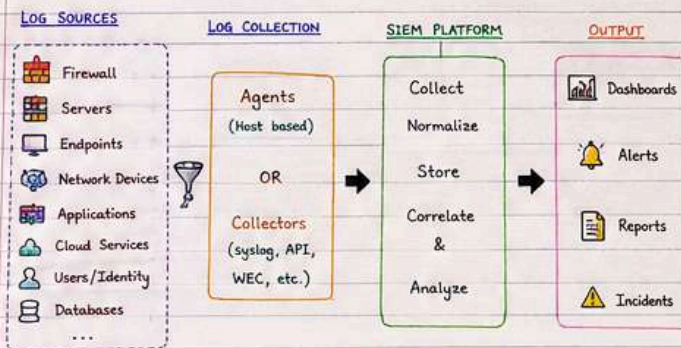
FEATURE	DESCRIPTION
Log Collection	Collects logs from multiple sources (agent, syslog, API, etc.)
Log Normalization	Converts logs into a common format
Log Storage	Stores logs for short-term & long-term retention
Correlation Engine	Combines events to detect suspicious or malicious activity
Real-time Alerting	Generates alerts based on rules & correlation
Dashboards	Provides real-time visibility using graphs, charts & widgets
Reporting	Custom & compliance reports
User & Entity Behavior Analytics	Detects anomalies in user/entity behavior
Integration	Works with other tools (EDR, Firewall, Threat Intel, SOAR, etc.)



REMEMBER:

- SIEM does not block attacks. It detects, alerts and helps us respond quickly.
- Prevention is done by Firewalls, EDR, IAM, etc.

BIG PICTURE: HOW SIEM WORKS



(1) TYPES OF LOG SOURCES

- Network Devices → Firewalls, Routers, Switches, IDS/IPS
- Endpoints → Windows, Linux, macOS (via agents)
- Servers → Web servers, File servers, Mail servers
- Applications → Web apps, Databases, ERP, CRM
- Cloud Services → AWS, Azure, O365, GCP
- Identity Sources → Active Directory, LDAP, RADIUS
- Security Tools → AV, EDR, DLP, Sandbox, WAF
- Others → Printers, IoT devices, VPN, Proxy, etc.

EXAMPLE:

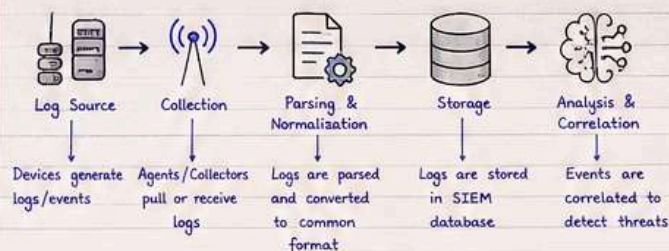
- Firewall Log : 2025-05-21 10:15:23 DENY 203.0.113.5 → 10.0.0.10
- Windows Log : 2025-05-21 10:15:24 User 'john' failed login
- EDR Log : 2025-05-21 10:15:25 suspicious process 'cmd.exe'
- AD Log : 2025-05-21 10:15:26 john logged in from WS-17

(2) LIMITATIONS OF WORKING WITH ISOLATED LOGS

- ✗ No full visibility → hard to see the complete attack chain
- ✗ Time consuming → logs are in different systems
- ✗ Inconsistent formats → hard to compare & analyze
- ✗ Storage limitations → logs rotate & get deleted
- ✗ Missed correlations → attacker actions across systems go unnoticed
- ✗ No central alerting → critical events may be delayed

Page 4.

(5) LOG INGESTION PROCESS IN SIEM



(6) ALERTING & ALERT ANALYSIS PROCESS



EXAMPLE ALERT FLOW

- Brute Force Login Detected.
- Rule: 5 failed logins from same IP within 5 mins
- Alert: High Severity
- Evidence: 203.0.113.5 tried 8 times to login to user 'admin'
- Action: Analyst checks logs, validates, escalates to L2.

WHEN TO ESCALATE?

- Alert is True Positive
- High/Critical Severity
- Incident affects business critical systems
- Requires advanced forensics
- Needs action beyond L1 scope

KEY TAKEAWAY

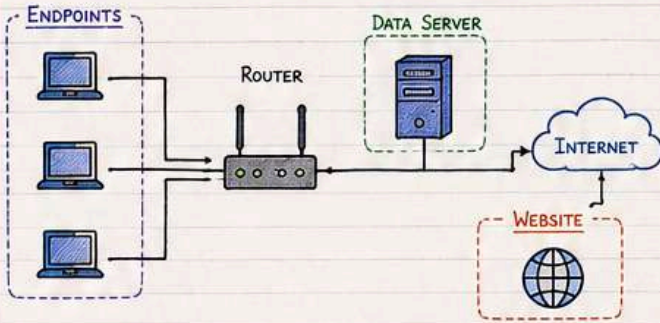
- SIEM collects everything, connects the dots and helps us see what attackers are trying to hide.



LOGS EVERYWHERE, ANSWERS NOWHERE

LOGS EVERYWHERE

Multiple devices in a network communicate with each other and, most of the time, with the Internet through a router. The image below shows an example of a simple network that comprises multiple Linux/Windows-based Endpoints, one data server, and one website.



These devices continuously generate logs of the activities that occur within them. We can also call these devices **log sources**. The logs they generate serve as a trail of all the activities and are extremely helpful for identifying malicious activities or general troubleshooting.

These log sources are mainly divided into two categories, which are discussed below.

(1) HOST-CENTRIC LOG SOURCES

These log sources capture events that occurred within or related to the host. Devices that generate host-centric logs include Windows, Linux, servers, etc.

Some examples of **host-centric logs** are:

- A user accessing a file
- A user attempting to authenticate.
- A process execution activity
- A process adding/editing/deleting a registry key or value.
- PowerShell execution



(2) NETWORK-CENTRIC LOG SOURCES

Network-related logs are generated when the hosts communicate with each other or access the internet to visit a website. Devices that generate **network-centric logs** are firewalls, IDS/IPS, routers, etc.

Some examples of **network-centric logs** are:

- SSH connection
- A file being accessed via FTP
- Web traffic
- A user accessing the company's resources through VPN.
- Network file sharing Activity



REMEMBER

Together, these host-centric and network-centric log sources constantly create numerous logs in a network.

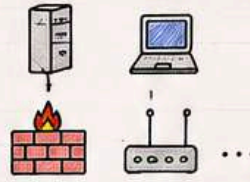


ANSWERS NOWHERE

Until now, it seems pretty straightforward that these log sources generate logs, we analyze them, and identify malicious activities. However, it's not that simple.

It has some **challenges**. Some of them are discussed below:

(1) NUMEROUS LOG SOURCES



A network has many log sources, which generate **hundreds of events per second**.

These logs are scattered across different devices, and examining the logs on each device one by one in case of an incident can be **tedious**.

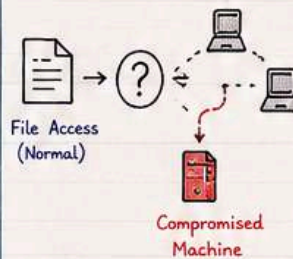
(2) NO CENTRALIZATION



As logs reside on the machines on which they are generated, you may need to connect with each log source via SSH, RDP, etc., to analyze logs from multiple log sources.

This is very **inefficient** and can **waste** a lot of your valuable time during the investigations.

(3) LIMITED CONTEXT

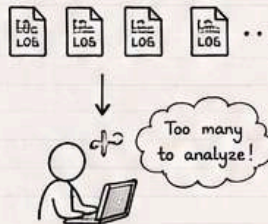


Individual logs cannot tell the whole story of an activity. During any incident, the individual activities on different log sources may seem harmless. But if these logs are correlated, they can indicate a whole different story.

Example:

A file access event is normal. But if correlated, we may find it was accessed after a lateral movement from a compromised machine!

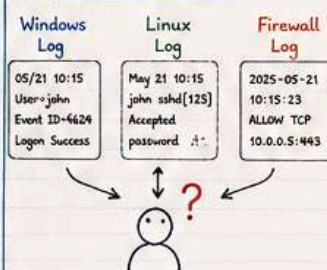
(4) LIMITED ANALYSIS



The log sources generate numerous logs per second, and analyzing all the logs from all the devices manually to identify any abnormal activity is nearly **impossible** for humans.

Realistically, the analysts will miss a lot of important logs in between the analyses due to their huge number.

(5) FORMAT ISSUES



Different log sources generate logs in various formats. Analysts need to know all these formats to analyze them, which can be extremely difficult, especially when dealing with numerous log sources in a network.



In the next task, we will learn about a **powerful technology** that can solve all these problems.

TASK 3 : WHY SIEM?

Page 1

1. THE NEED

In the previous task, we saw that multiple devices in a network generate numerous logs of various types and that analyzing them individually is difficult and inefficient.

So, how can we more efficiently manage this flood of data and extract valuable results?

This is where **SIEM** comes into play.

2. WHAT IS SIEM?

SIEM (Security Information and Event Management) is a security solution that **collects logs** from various types of log sources, **standardizes** their format into a consistent one, **correlates** them, and **detects malicious activities** using detection rules.

SIEM =
Better Visibility
Faster Detection
Stronger Security

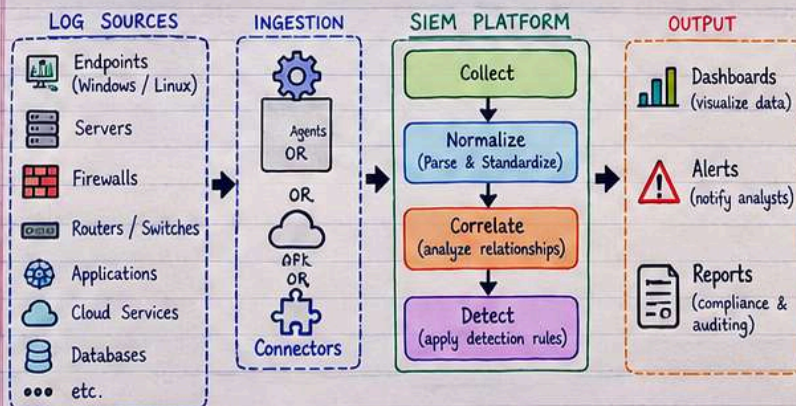
3. WHY SIEM MATTERS?

- ✓ Solves the problems of scattered and unstructured logs.
- ✓ Helps detect threats in real time.
- ✓ Provides meaningful insights, not just raw data.
- ✓ Makes incident investigation faster and more effective.
- ✓ Supports compliance and auditing needs.

REMEMBER

SIEM turns a
"pile of logs"
into
"useful intelligence."

4. HOW SIEM WORKS (BIG PICTURE)



Once in the SIEM, the flood of logs becomes organized, correlated and turned into actionable insights!

5. KEY FEATURES OF SIEM

Let's explore the core features that make a SIEM so powerful.

1 Centralized Log Collection



- Collects logs from all sources (endpoints, servers, firewalls, etc.)
- Pulls logs through lightweight agents or APIs
- Stores everything in one place.
- No need to log in to each machine individually

Benefit

Saves time and makes log analysis efficient.

2 Normalization of Logs



- Raw logs are in different formats and structures.
- SIEM parses and breaks logs into fields (parsing).
- Converts all logs into a consistent format (normalization).
- Makes logs easier to search, query and analyze.

Example

Windows log → dest_ip
Linux log → user
Firewall log → src.port
All normalized into same format

3 Correlation of Logs



- Finds relationships between events from different sources.
- Helps identify complex threats and attack patterns.
- Less effective on isolated, individual logs.

Real-life Example

1. Harris logs in via VPN from 1. a new IP (never used before)
 2. Accesses documents on a shared drive
 3. Executes a PowerShell script
 4. Outbound network connection
- Individually looks normal, but together may indicate data exfiltration from compromised VPN credentials.

4 Real-time Alerting



- Uses detection rules to analyze events.
- Triggers alerts when suspicious activity is detected.
- Analysts are notified and can investigate within the SIEM platform.

Analyst Action

1. Alert comes in
2. Investigate in SIEM
3. Validate the threat
4. Take further action (escalate, contain, etc.)

5 Dashboards and Reporting



- Presents data in visual formats (dashboards).
- Default and custom dashboards available.
- Helps in quick analysis and decision-making.

Common Info in Dashboards

- Alert highlights
- System notifications
- Health alert
- Failed login attempts
- Events ingested count
- Rules triggered
- Top domains visited

6. EXAMPLE: SPLUNK SIEM DASHBOARD



6 OTHER (ADVANCED) FEATURES

- Integration with threat intelligence feeds
- Extensive data retention
- Powerful searching capabilities
- Automated response (playbooks)
- Integrations with SOAR, tip tools, etc.

IN SHORT...

- ✓ Collect
- ✓ Normalize
- ✓ Correlate
- ✓ Detect
- ✓ Alert
- ✓ Analyze
- ✓ Gain actionable insights
- ✓ Stay ahead!

★ **SIEM = VISIBILITY + CONTEXT + ACTION = A STRONGER, SAFER NETWORK**

On to the next task where we will look at different log sources in detail →

Page 2.

LOG SOURCES AND INGESTION

(1) LOG SOURCES

Every device in the network generates some kind of log whenever an activity is performed on it, such as a user visiting a website, connecting to SSH, logging into their workstation, etc.

💡 These logs are extremely useful for security monitoring, troubleshooting and incident investigations.

Let's see what the logs of some common devices that are found in a network environment look like.

A) WINDOWS MACHINE

Windows records every event that can be viewed through the Event Viewer. It assigns a unique ID to each type of log activity, making it easy for the analyst to examine and keep track of.

- To view events in a Windows environment, type **Event Viewer** in the search bar. This takes you to the tool where different logs are stored and can be viewed, as shown below.
- These logs from all Windows endpoints are forwarded to the SIEM solution for monitoring and better visibility.

Shows Events in Event Viewer Interface

Event Viewer (Local)		Security Number of events: 1,254 (1) New events available				
Custom Views		Level	Date and Time	Source	Event ID	Task Category
Windows Logs		Information	5/28/2025 1:33:21 PM	Microsoft Windows security	4624	Logon
Application		Success Audit	5/28/2025 1:33:22 PM	Microsoft Windows security	4624	Logon
Security		Warning	5/28/2025 1:33:25 PM	Microsoft Windows security	4634	Logoff
Setup		Error	5/28/2025 1:33:30 PM	Service Control Manager	7031	None
System		Information	5/28/2025 1:33:31 PM	Microsoft Windows security	4672	Special Logon
Forwarded Events						
Applications and Services Logs						
Subscriptions						

★ Many types of logs: Application, Security, System, Setup, etc.

B) LINUX MACHINE

Linux OS stores all the related logs, such as events, errors, warnings, etc. These are then ingested into SIEM for continuous monitoring.

Common Log Locations:

- **/var/log/httpd**: Contains HTTP Request / Response and error logs.
- **/var/log/cron**: Events related to cron jobs are stored in this location.
- **/var/log/auth.log** and **/var/log/secure**: Stores authentication-related logs.
- **/var/log/kern**: This file stores kernel-related events.

Here is a sample of a cron log:

```
May 28 13:04:20 ebr crond[2843]: /usr/sbin/crond 4.4 dillon's
cron daemon, started with loglevel notice
May 28 13:04:20 ebr crond[2843]: no timestamp found (user
root job sys-hourly)
May 28 13:04:20 ebr crond[2843]: no timestamp found (user
root job sys-daily)
May 28 13:04:20 ebr crond[2843]: no timestamp found (user
root job sys-weekly)
May 28 13:04:20 ebr crond[2843]: no timestamp found (user
root job sys-monthly)
Jun 13 07:46:22 ebr crond[3592]: unable to exec /usr/sbin/
sendmail: cron output for user root job sys-daily
to /dev/null
```

★ REMEMBER

- ✓ Logs are the digital footprints of every activity.
- ✓ Without logs → No visibility.
- ✓ Without visibility → No security!



C) WEB SERVER

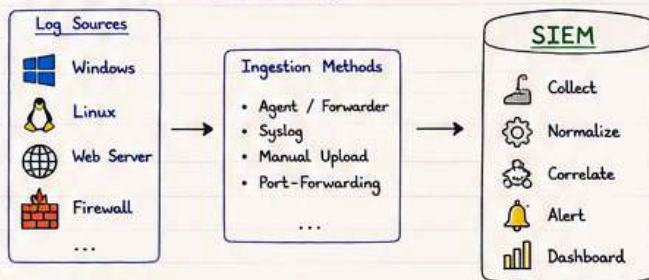
It is important to monitor all requests/responses coming in and out of the web server for any potential web attack attempt. In Linux, common locations to write all apache-related logs are **/var/log/apache** or **/var/log/httpd**.

Here is an example of Apache Logs:

```
192.168.21.200 - - [21/March/2022:10:17:10 -0300]
"GET /cgi-bin/try HTTP/1.0" 200 3395 "-"
"Mozilla/5.0 (Windows NT 10.0; Win64; x64)
AppleWebKit/537.36 (KHTML, like Gecko)
Chrome/98.0.4758.102 Safari/537.36"

127.0.0.1 - - [21/March/2022:10:22:04 -0300]
"GET / HTTP/1.0" 200 2216 "-" "curl/7.68.0"
```

LOG INGESTION IN SIEM (High Level Flow)



(2) LOG INGESTION

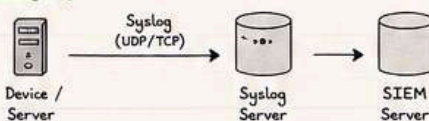
All these logs provide a wealth of information and can help identify security issues. Each SIEM solution has its own way of ingesting the logs. Some common methods used by these SIEM solutions are explained below:

1. Agent / Forwarder



These SIEM solutions provide a lightweight tool called an agent (forwarder by Splunk) that gets installed on the Endpoint. It is configured to capture and send all the important logs to the SIEM server.

2. Syslog



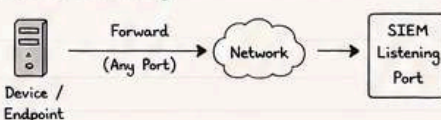
Syslog is a widely used protocol to collect data from various systems like web servers, databases, etc., and send real-time data to the centralized destination.

3. Manual Upload



Some SIEM solutions, like Splunk, ELK, etc., allow users to ingest offline data for quick analysis. Once the data is ingested, it is normalized and made available for analysis.

4. Port-Forwarding



SIEM solutions can also be configured to listen on a certain port, and then the endpoints forward the data to the SIEM instance on the listening port.

DATA INGESTION OPTIONS IN SPLUNK SIEM

splunk>enterprise Apps Messages Settings Activity Help Find

Add Data

How do you want to add data?

Upload
Upload a file from my computer. (.log, .csv, etc.)

Monitor
Continuously monitor a location.

Forward
Forward data from a remote host.

Network
Listen on a network port.

Scripts
Run a script to collect data.

Help
Learn more about adding data.

💡 Log sources → Ingestion → SIEM → Intelligence → Security
Collect • Normalize • Correlate • Detect • Respond

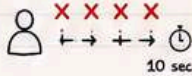
ALERTING PROCESS AND ANALYSIS

1) BEHIND THE TRIGGERED ALERTS

We learned that a SIEM solution detects threats by correlating logs from the log sources and triggers alerts, but do we know the **magic** behind these detections?

SIEM solution has **DETECTION RULES** that catch threats. These rules play an important role in the timely detection of threats, allowing analysts to **take action on time**.

Detection rules are pretty much logical expressions set to be triggered. A few examples of detection rules are:

- If a user gets five failed Login Attempts in 10 seconds, raise an alert for Multiple Failed Login Attempts 
- If login is successful after multiple failed login attempts, raise an alert for Successful Login After multiple Login Attempts 
- A rule is set to alert every time a user plugs in a USB (Useful if USB is restricted as per the company policy) 
- If outbound traffic is > 25 MB, raise an alert to potential data exfiltration Attempt (Usually, it depends on the company policy) 

2) HOW IS A DETECTION RULE CREATED?

To explain how the rule works, consider the following Eventlog use cases:

USE-CASE 1:

Adversaries tend to remove the logs during the post-exploitation phase to remove their tracks. A unique Event ID **104** is logged every time a user tries to remove or clear event logs. To create a rule based on this activity, we can set the condition as follows:

RULE: If the Log source is WinEventLog AND EventID is **104** - Trigger an alert **Event Log Cleared**



USE-CASE 2:

Adversaries use commands like **whoami** after the exploitation/privilege escalation phase. The following fields will be helpful to include in the rule.

- Log source: Identify the log source capturing the event logs
- Event ID: Which Event ID is associated with Process Execution activity? In this case, Event ID **4688** will be helpful.
- NewProcessName: Which process name will be helpful to include in the rule?

RULE: If Log Source is WinEventLog AND EventCode is **4688**, and NewProcessName contains **whoami**, then Trigger an **ALERT WHOAMI command Execution DETECTED**



 In the previous task, the importance of field-value pairs was discussed. Detection rules keep an eye on the values of certain fields to get triggered. That is the reason why it is important to have **normalized** logs ingested.

3) ALERT INVESTIGATION

When monitoring SIEM, analysts spend most of their time on dashboards, as they display various key details about the network in a very summarized way.



ACTIONS AFTER ANALYSIS

- ⊗ Alert is a False Positive. → May require tuning the rule to avoid similar False positives from occurring again.
- ✓ Alert is a True Positive. → Perform further investigation.
- 📞 Contact the asset owner → to inquire about the activity.
- 🛡️ Suspicious activity is confirmed. → Isolate the infected host.
- 🚫 Block the suspicious IP. → Add the IP to firewall/IDS blocklist.



Investigate
Analyze
Respond
Protect

★ KEY TAKEAWAYS

- ✓ SIEM uses detection rules to find suspicious or malicious activities.
- ✓ Rules are built using field-value pairs from normalized logs.
- ✓ Alerts help analysts act quickly and reduce the impact of threats.
- ✓ Investigation + Analysis = Better Decisions & Stronger Security

Remember :

Good rules + Good data (normalized logs)
= Accurate alerts + Effective security operations.

CHAPTER SUMMARY / CONCLUSION

From Logs to Security Intelligence



Remember:

You can't protect what you can't see. SIEM gives you the visibility to protect.



In this chapter, we learned how every activity in a network leaves a digital footprint in the form of logs, and how a SIEM solution turns those raw logs into actionable security insights through collection, normalization, correlation, alerting and analysis.

① LOGS EVERYWHERE

Every device generates logs.

- **Host-Centric Logs** - events within the host (e.g., logins, file access, process execution, registry changes, PowerShell execution)
- **Network-Centric Logs** - events related to communication between hosts or with Internet (e.g., SSH, FTP, Web traffic, VPN, file sharing)

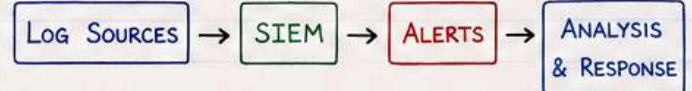


② CHALLENGES WITH LOGS

- ✓ Numerous log sources → huge volume
- ✓ No centralization → hard to access & slow
- ✓ Limited context → incomplete picture
- ✓ Limited analysis → humans can't analyze everything
- ✓ Format issues → different formats, hard to understand

③ WHY SIEM?

SIEM collects, normalizes, correlates logs and detects threats using detection rules.



SIEM = Centralized Visibility + Early Detection + Faster Response

④ FEATURES OF SIEM

- ★ Centralized Log Collection → all logs in one place
- ★ Normalization of Logs → convert to consistent format
- ★ Correlation of Logs → connect the dots
- ★ Real-time Alerting → detect & notify in real time
- ★ Dashboards & Reporting → actionable insights at a glance (+ many more: threat intel feeds, retention, search, etc.)

⑤ LOG SOURCES & INGESTION

- Windows Logs
- Linux Logs
- Web Server Logs
- Network Device Logs
- ...

Ingestion Methods

- Agent / Forwarder
- Syslog
- Manual Upload
- Port-Forwarding

⑥ ALERTING PROCESS

1) Detection Rules

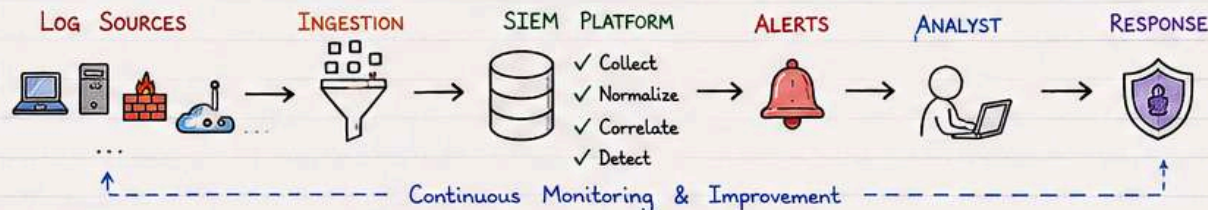
- Logical conditions on field-value pairs
- Examples: Failed logins, Event ID 104, whoami execution, high outbound traffic, etc.

2) Alert Investigation

- Analyze events, check rule conditions, determine True/False Positive
- Take actions → escalate, isolate host, block IP, tune rules, etc.



THE BIG PICTURE (HOW IT ALL WORKS)



KEY TAKEAWAYS

- ✓ Logs are everywhere.
- ✓ Raw logs are hard to analyze.
- ✓ SIEM turns data into intelligence.
- ✓ Detection rules are the heart of alerting.
- ✓ Timely investigation leads to effective response.



IN SHORT...

SIEM is the backbone of a SOC. It gives us visibility, context, and the ability to detect and respond to threats faster and smarter.



REMEMBER THIS MANTRA:

Collect → Normalize → Correlate → Detect → Alert → Analyze → Respond

= Stronger Security Posture